

COMPUTATIONAL TIME MACHINE

From Automated AI Researcher to Accelerated AI Civilization

A systems research proposal for compressing validated technological R&D time

Core idea: do not merely scale the number of AI researchers. Build a persistent, generational R&D ecosystem in which validated discoveries change the tools and capabilities available to successor generations.

THE STEP BEYOND PARALLEL AGENTS

Recent frontier systems show that large coordinating-agent populations can explore many research paths in parallel. CTM asks the next question: can independently verified discoveries persist across research generations, alter the research substrate, and produce cumulative acceleration rather than one-shot problem solving?

CIVILIZATION-IN-THE-LOOP (CitL)

- Specialized AI populations: scientists, engineers, critics, replication teams, tool builders, and resource allocators.
- Persistent provenance: papers, code, proof objects, experiment logs, datasets, failures, and causal dependency graphs.
- A generation compiler: verified advances can become new tools, workflows, models, simulators, or experimental capabilities for the next epoch.
- Physical closure: robotic labs and automated test rigs return real measurements when simulation is insufficient.

CORE LOOP

Research -> Verify -> Inherit -> Improve the environment -> Next generation -> Repeat

FIRST DEFENSIBLE EXPERIMENT

Run a 30-day, sandboxed study in a closed, high-verifiability domain. Compare two matched programs under the same total compute and initial tools: (A) conventional parallel agentic R&D; (B) generational CitL R&D, where only independently validated discoveries are inherited. Success requires a reproducible chain of novel advances in which later results causally depend on earlier machine-generated discoveries - not merely more tokens, attempts, or duplicated work.

WHY NOW

- OpenAI reports a trajectory from an automated research intern toward an automated AI researcher under human supervision.
- By mid-August 2026, OpenAI reports 3.1 agent-workdays per human workday across its research organization.
- The September 2026 Navier-Stokes effort used a coordinating-agent system with on the order of 10,000 concurrent agents and cross-pollination of intermediate results.
- OpenAI explicitly identifies research orchestration, sandboxed runtimes, feedback loops, and RSI measurement as active research areas.

KEY METRIC

TTCR

Technological Time Compression Ratio
= rate of independently validated frontier advancement under CTM / rate under a matched conventional R&D baseline.

SAFETY PREMISE

Generational transitions are supervised. Capability-changing artifacts are candidates for promotion, not automatically inherited. Safety evaluation, provenance review, sandboxing, rollback, and human authorization remain explicit gates.

Research question: Can a verified generational AI R&D ecosystem make the compression factor itself improve over successive generations?

Context: OpenAI, "Research acceleration: The view inside OpenAI" (Sep 6, 2026); "On the Navier-Stokes Millennium Prize Problem" (Sep 8, 2026); OpenAI RSI research role; OpenAI policy note on recursive self-improvement. openai.com/research